# AI-Based Multi-Agent Framework for Emergency Response Coordination and Smart Traffic Clearance in Urban Environments

*NLP Severity Classification, Gradient Boosting ETA Prediction, and Rule-Based Multi-Agent Coordination*


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***Abstract—****Delays in emergency medical response are a major determinant of preventable mortality and morbidity in dense urban environments, where ambulance dispatch, hospital selection, traffic signal coordination, and police deployment are typically handled as independent, manually coordinated processes. This paper proposes an AI-based multi-agent framework for emergency response and smart traffic clearance, developed as a simulation-driven smart-city platform for Bengaluru. The system integrates five coordinated agents — a natural language processing (NLP) severity classification agent using TF-IDF features with logistic regression, a Gradient Boosting-based ambulance ETA prediction agent, a Random Forest-based trauma-aware hospital optimization agent, a Random Forest-based traffic congestion and green-corridor signal agent, and a severity-driven police deployment agent — orchestrated through a central multi-agent coordinator that resolves inter-agent conflicts such as ambulance-police routing overlaps and hospital capacity contention. The framework is deployed as a three-tier web application comprising an HTML5/Leaflet.js front end, a FastAPI orchestration and WebSocket streaming backend, and independent scikit-learn based machine learning microservices, with live visualization on an OpenStreetMap-based dispatch dashboard. The system is evaluated on a curated dataset of Bengaluru hospitals, ambulance base locations, and simulated emergency scenarios. Preliminary benchmarking indicates trauma-appropriate hospital selection in five of eight tested scenarios, an ETA prediction RMSE of 1.28 minutes, a hospital selection Q-value RMSE of 0.037, and successful multi-agent conflict resolution across all test scenarios. Taken together, the system design and benchmark results support the feasibility of a modular, interpretable multi-agent architecture for coordinated emergency response, offering a foundation for future integration with live GPS tracking, IoT-based traffic infrastructure, and municipal emergency dispatch systems.*



***Index Terms—****Emergency Response, Multi-Agent Systems, Ambulance Dispatch, ETA Prediction, Hospital Optimization, Smart Traffic Signal Control, Green Corridor, NLP Severity Classification, Gradient Boosting, Random Forest, Smart City.*


## I. INTRODUCTION

Emergency medical services (EMS) operate under strict time constraints in which every additional minute of ambulance response time measurably reduces the probability of favourable patient outcomes, particularly for trauma, cardiac, and stroke cases. In large Indian metropolitan areas such as Bengaluru, ambulance response is further degraded by unpredictable traffic congestion, ad hoc hospital selection, and a lack of coordination between emergency medical, traffic, and police authorities. Existing dispatch practice largely relies on static protocols — nearest-hospital routing, fixed-time traffic signals, and manually radioed police support — none of which adapt dynamically to real-time incident severity, traffic state, or hospital capacity.

Recent advances in machine learning and multi-agent systems offer an opportunity to automate and optimize each stage of the emergency response pipeline: natural language processing can extract severity signals from free-text incident descriptions; regression-based models can predict ambulance arrival time from historical trip data; classification and optimization models can select trauma-appropriate hospitals under capacity constraints; and reinforcement-learning-adjacent heuristics can coordinate traffic signal preemption and police unit allocation. However, most prior systems address these sub-problems in isolation, without a unifying coordination layer capable of resolving conflicts that arise when multiple agents act on overlapping urban resources.

This paper presents an AI Emergency Response & Smart Traffic Clearance System that unifies five coordinated AI agents — severity classification, ETA prediction, hospital optimization, traffic signal control, and police deployment — under a central multi-agent coordinator, deployed as a three-tier simulation platform for Bengaluru. The specific contributions of this work are:

• An NLP-driven severity classification pipeline that converts free-text incident reports into an actionable urgency score.

• A Gradient Boosting ETA predictor and Random Forest-based trauma-aware hospital optimizer that jointly determine ambulance destination.

• A Random Forest traffic congestion model that drives a simulated green-corridor signal-clearance mechanism.

• A rule-based multi-agent coordinator that resolves ambulance-police routing conflicts and hospital capacity contention.

• A real-time WebSocket-driven dispatch dashboard for end-to-end visualization of the emergency response workflow.

## II. LITERATURE REVIEW

Prior work relevant to this system spans four areas: NLP-based incident severity classification, machine-learning-based ambulance ETA and demand prediction, hospital and destination selection optimization, smart traffic signal preemption, and multi-agent coordination for emergency response.

### *A. Severity Classification from Incident Text*

Natural language processing has been widely applied to classify severity and route incident reports from unstructured text. Wali et al. [1] combined predictive text analytics with heterogeneity-based statistical modelling to analyse injury severity in pedestrian and bicyclist crash narratives, while Wan et al. [2] demonstrated that NLP-processed social media data can empower real-time traffic incident reporting systems. More recent work has explored large language model-derived features for severity classification of traffic incident reports, showing that ensemble classifiers such as Random Forest and XGBoost benefit substantially from LLM-extracted textual features over structured fields alone [3]. In the clinical domain, deep continual learning approaches have been used to prioritize emergency medical call incidents by life-threatening level under dataset shift [4], and systematic reviews confirm that a broad range of NLP techniques achieve favourable performance for incident and adverse-event severity classification from free text [5]. These findings motivate the use of a lightweight TF-IDF plus logistic regression pipeline for real-time severity scoring in the proposed system, prioritizing interpretability and inference speed over the heavier LLM-based approaches used in offline analytics settings.

### *B. Ambulance ETA and Demand Prediction*

Gradient-boosted tree methods have become the dominant approach for ambulance response-time and demand forecasting. Industry deployments in the Netherlands have applied LightGBM to next-day ambulance demand forecasting to help EMS regions meet national response-time targets [6], while a two-stage ensemble integrating Random Forest, Support Vector Regression, and XGBoost has been shown to outperform single-model baselines for monthly ambulance demand forecasting across sixteen districts in Thailand [7]. Yadav et al. [8] compared a custom large numerical model against XGBoost for EMS response-time prediction, confirming gradient boosting as a strong baseline for this task. These results support the choice of a Gradient Boosting Regressor as the ETA prediction agent in the proposed framework.

### *C. Hospital Selection and Destination Optimization*

Destination hospital selection has been modelled using both learned and optimization-based approaches. Ambulance dispatch and casualty distribution during mass-casualty incidents has been formulated as a Markov Decision Process solved via deep-neural-network-based approximate policy iteration to jointly optimize dispatch, redeployment, and hospital selection [9]. Two-stage routing algorithms that first identify the nearest capable facility and then verify service availability have been proposed for EMS availability-aware routing [10], and empirical studies of city-scale ambulance dispatch note that the widely assumed nearest-hospital principle is violated in over 40% of real transports due to capacity, specialization, and patient-specific factors [11]. Deep reinforcement learning using Proximal Policy Optimization has also been applied to recommend optimal destination hospitals under multi-objective reward functions balancing survival probability, transport duration, and facility capacity [12]. The proposed system adopts a Random Forest Regressor as a lightweight, interpretable analogue to these approaches, scoring candidate hospitals against trauma capability, distance, and estimated capacity.

### *D. Smart Traffic Signal Preemption and Green Corridors*

IoT-enabled green-corridor systems dynamically override fixed-time traffic signal patterns to create an unimpeded path for emergency vehicles. RFID-based vehicle detection combined with microcontroller-driven signal preemption has been demonstrated and validated through combined SUMO and NS2 simulation [13], while UWB-based emergency vehicle identification has been proposed for dynamic green-corridor activation with simultaneous notification of nearby vehicles to clear lanes [14]. Reviews of IoT-based emergency vehicle services in intelligent transportation systems summarize complementary preemption strategies, including central-controller phase overrides and Petri-net-based preemption models [15]. These works motivate the traffic prediction agent's design in the proposed system, which uses a Random Forest classifier over simulated congestion features to select an optimal signal corridor mode.

### *E. Multi-Agent Coordination for Emergency Response*

Coordinating multiple autonomous agents across a shared urban environment is a well-studied challenge in emergency response research. Sivagnanam et al. [16] proposed a hierarchical multi-agent reinforcement learning approach for emergency responder stationing that reduces per-decision computation time by three orders of magnitude relative to Monte Carlo tree search baselines while improving average response time, evaluated on real dispatch data from Nashville and Seattle. Pettet et al. [17] examined algorithmic decision procedures for emergency response in smart and connected communities, and a recent multi-agent reinforcement learning framework has been proposed to optimize smart cities as systems-of-systems spanning transportation, public safety, and communication domains under decentralized policy learning [18]. Related coordination challenges also arise in police resource deployment: continuous-time crime models have been used to optimize patrol allocation [19], mixed-integer optimization under uncertainty has been applied to daily law-enforcement deployment planning to minimize response-time risk [20], and multi-agent reinforcement learning has more recently been used to optimize cooperative patrol routing for urban crime surveillance [21]. The proposed system's rule-based multi-agent coordinator draws on these principles of conflict resolution and priority-based allocation, applying them to ambulance-police routing conflicts and hospital capacity contention in a real-time simulation setting.

## III. PROBLEM STATEMENT

Existing emergency response workflows in urban Indian settings treat ambulance dispatch, hospital selection, traffic clearance, and police deployment as loosely coupled, largely manual processes, resulting in avoidable delay accumulation across the response chain. A system that predicts severity directly from an unstructured incident description, estimates ambulance arrival time under current network conditions, selects a trauma-appropriate hospital under capacity constraints, activates a traffic green corridor, and allocates police support — all coordinated through a shared decision layer that resolves resource conflicts — has the potential to substantially reduce end-to-end emergency response latency. The specific objectives of this project are: to design an NLP-based severity classification agent operating on free-text incident reports; to implement a Gradient Boosting ETA prediction agent and a Random Forest-based trauma-aware hospital optimization agent; to implement a Random Forest-based traffic congestion agent driving simulated signal preemption; to design a rule-based multi-agent coordinator capable of resolving ambulance-police and hospital-capacity conflicts; and to deploy the complete pipeline as a real-time, WebSocket-driven dispatch simulation for the city of Bengaluru, benchmarked against curated hospital, ambulance base, and incident datasets.

## IV. METHODOLOGY

The proposed framework processes an incoming emergency report through five coordinated agents, arbitrated by a central coordinator, before rendering results on a live dispatch dashboard.

*1) NLP Severity Classification Agent:*

Incoming free-text incident descriptions (e.g., “major accident, multiple injuries”) are vectorized using term frequency–inverse document frequency (TF-IDF) representations and classified by a logistic regression model into discrete severity tiers. This lightweight design favours millisecond-scale inference suitable for real-time SOS triggering over heavier transformer-based classifiers, consistent with the interpretability requirements identified in prior incident-classification literature [3], [5].

*2) ETA Prediction Agent:*

Ambulance estimated time of arrival is predicted using a Gradient Boosting Regressor trained on features including base-to-incident distance, historical traffic conditions, and time-of-day effects, following the gradient-boosting formulation widely validated for EMS response-time prediction [6]–[8]. The predicted ETA is subsequently used both for hospital-selection scoring and for dispatch dashboard display.

*3) Hospital Optimization Agent:*

Candidate hospitals are scored using a Random Forest Regressor over trauma-capability match, distance, predicted travel time, and estimated bed/resource availability drawn from the curated Bengaluru hospital dataset. This departs from the static nearest-hospital heuristic, in line with evidence that a substantial fraction of real-world transports are not routed to the nearest facility [11], and is designed as an interpretable analogue to the MDP- and DRL-based hospital selection strategies proposed in prior mass-casualty and city-scale dispatch research [9], [12].

*4) Traffic Prediction and Green Corridor Agent:*

A Random Forest classifier estimates congestion state along candidate ambulance routes and selects a signal corridor mode, simulating the RFID/UWB-triggered green-corridor preemption strategies documented in the IoT-based traffic management literature [13], [14]. In the current simulation, corridor activation is represented as a signal-state override rather than a live hardware integration, reflecting the platform's simulation-driven scope.

*5) Police Deployment Agent:*

Police unit allocation is determined by incident severity and predicted route risk, drawing on the resource-allocation principles established in patrol optimization literature [19], [20], and is dispatched in parallel with the ambulance and traffic agents rather than sequentially.

*6) Multi-Agent Coordinator:*

A central coordinator arbitrates the outputs of the five agents, resolving conflicts including ambulance-versus-police routing overlaps, hospital bed-availability contention, traffic-corridor rerouting, and priority-based sequencing of concurrent emergencies. The coordinator's conflict-resolution role is analogous to the hierarchical and decentralized coordination mechanisms studied in recent multi-agent emergency-response and smart-city literature [16]–[18], applied here as a deterministic rule-based layer suited to the real-time constraints of the simulation platform.

## V. SYSTEM ARCHITECTURE

The system follows a three-layer architecture: a frontend layer, a FastAPI backend hosting the multi-agent coordinator, and a machine learning and data layer.

| **Layer** | **Component** |
|---|---|
| Frontend (HTML5 / Leaflet.js) | Live map, dispatch console, results dashboard |
| Backend (FastAPI) | REST APIs, WebSocket streaming, multi-agent coordinator |

| | |
|---|---|
| Agent Layer (scikit-learn) | NLP severity, ETA (Gradient Boosting), hospital (Random Forest), traffic (Random Forest), police agents |
| Data Layer | Curated hospital/ambulance CSVs, GeoJSON, OpenStreetMap tiles |
| Output Layer | Real-time dispatch analytics and ML metrics dashboard |

TABLE I. SYSTEM ARCHITECTURE OVERVIEW

The frontend, implemented in HTML5, CSS3, JavaScript and Leaflet.js over OpenStreetMap tiles, submits incident reports to a FastAPI backend, which orchestrates sequential and parallel agent execution and streams dispatch events over a WebSocket endpoint (/ws/dispatch) to the live map dashboard. The five agents — ambulance, traffic, hospital, police, and the coordinator itself — are implemented as independent Python modules under a common agents package, each interfacing with its corresponding scikit-learn model in the ml directory. Optional integration with the OpenRouteService API supports realistic route geometry generation.

## VI. EXPECTED / PRELIMINARY OUTCOMES

Benchmark testing on curated Bengaluru hospital and ambulance-base datasets indicates trauma-appropriate hospital selection in five of eight tested emergency scenarios, an ETA prediction RMSE of 1.28 minutes, and a hospital Q-value RMSE of 0.037. The multi-agent coordinator successfully resolved routing and resource conflicts across all tested scenarios, confirming the feasibility of the rule-based coordination layer under the simulated conditions evaluated. These results are broadly consistent with the response-time and destination-optimization gains reported in prior gradient-boosting and Random Forest-based EMS studies [6]–[8], [11], though direct comparison is limited by differences in dataset scale and geography. Future evaluation is expected to extend benchmarking to a larger set of simulated incidents and to compare the Random Forest hospital optimizer against the MDP and DRL-based destination-selection baselines discussed in Section II-C [9], [12].

## VII. CONCLUSION

This paper has presented an AI-based multi-agent framework for emergency response coordination and smart traffic clearance, combining NLP-based severity classification, Gradient Boosting ETA prediction, Random Forest-based hospital and traffic optimization, and a rule-based multi-agent coordinator within a three-tier simulation platform for Bengaluru. The system addresses the fragmentation of existing manual emergency response workflows by unifying ambulance dispatch, hospital selection, traffic signal clearance, and police deployment under a shared coordination layer, evaluated through a live map-based dispatch simulation with WebSocket streaming.

Future work will focus on integrating live BBMP hospital and GPS ambulance tracking APIs, incorporating IoT-based real traffic signal control in place of the current simulated green-corridor mechanism, applying reinforcement learning to the route-optimization and multi-agent coordination layers as demonstrated in recent hierarchical MARL emergency-response research [16], [18], extending CCTV-based automated accident detection, and evaluating cloud deployment and multi-city expansion across Karnataka.

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